

U30CAM-4K-S1 UserManual



1. General Description:

The Innomaker USB3.0 UVC Camera Module (IMX415) is a high-performance photographic solution designed for multi-platform applications. This module combines advanced imaging technology with excellent compatibility to provide users with a clear and smooth imaging experience.

This Innomaker USB3.0 UVC Camera Module is equipped with a USB 3.0 high-speed interface and a high-performance ISP (Image Signal Processor), ensuring extreme speed in data transmission while providing optimized image enhancement functions for exceptional picture quality. It boasts outstanding cross-platform compatibility, seamlessly supporting Windows 11/10/7, macOS, Linux, Ubuntu, Android, Raspberry Pi, Jetson Nano, and various ARM development boards. Its Plug & Play design, coupled with native UVC drivers, allows users to quickly get started without the need for additional driver installations.

In terms of imaging performance, this module offers video resolutions up to 4K@30fps and 1080P@60fps, supporting both YUY2 and MJPEG image formats. Multiple resolution options cater to diverse application scenarios requiring specific image quality and fluidity. Furthermore, it features a wide field of view with $Fov(D) = 116$ degrees and $Fov(H) = 105$ degrees, capable of capturing a broader picture, making it suitable for situations requiring wide-range monitoring or video conferencing. To ensure stability, the module has a built-in common mode filter, providing ESD (electrostatic discharge) and EMI (electromagnetic interference) protection for high-speed differential signal lines. This design ensures stable performance in various complex environments, effectively reducing signal noise and enhancing system reliability.

The Innomaker USB3.0 UVC Camera Module, with its excellent performance and wide compatibility, is suitable for a variety of innovative applications. For instance, in embedded vision systems, it is ideal for Raspberry Pi, Jetson Nano, and other ARM development boards, enabling the development of embedded applications such as smart access control, robot vision, and drone image acquisition. For video conferencing and live streaming, its high resolution and wide-angle view make it an ideal choice for high-quality video conferences, online education, or live streaming, providing clear and smooth visuals. In the field of industrial inspection and monitoring, it can be used for product quality inspection and production line monitoring, with its stability and anti-interference capabilities ensuring data accuracy. Additionally, it can be integrated into smart home and security systems to achieve real-time monitoring, facial recognition, and other functions, enhancing home security levels. In medical imaging equipment, it can also be used to assist diagnostic devices by providing clear image data.

2. Features

1. [USB3.0 Device] Equipped with a USB 3.0 high-speed interface and a high-performance ISP (Image Signal Processor), ensuring optimal image enhancement and exceptional picture quality.
2. [Wide USB3.0/2.0 Compatibility] Fully compatible with Windows 11/10/7, macOS, Linux, Ubuntu, Android, Raspberry Pi, ARM boards, laptops, and desktops. Plug & play functionality with native UVC driver.
3. [High-Definition Performance] Supports video resolutions up to 4K@30fps and 1080P@60fps. It also supports YUY2 and MJPEG formats, with multiple resolution options to suit various applications.
4. [Wide Angle] Fov(D) = 116 degrees and Fov(H) = 105 degree.
5. [Enhanced Protection] Built-in common mode filter provides ESD (electrostatic discharge) and EMI (electromagnetic interference) protection for high-speed differential signal lines, ensuring stable performance and reduced signal noise in various environments

3. Technical Specification

Sensor	Type	IMX415
	Size	1/2.8 Inch
	Resolution	3840*2160
	Image Color	Colours
	Shutter type	Rolling shutter
Lens	F(N) /Aperture	2.0
	Focal Length	2.25
	TV DISTORTION	<-6.2%
	CRA	5.4 Degree
	Lens Diameter	M12
	Lens Seat Spacing	20mm
	FOV-D	116Degree
	FOV-H	105 Degree
PCBA	Size	38*38 mm
	Layers	4 layers
	Mounting Holes	4 counts, Diameter 2.2mm
	Trigger	None
	Strobe Flash	None
	Night vision	None
	ESD/EMI	Yse
Interfaces	Communication Interface	USB 3.0 High-Speed
	Cable Length	1 M
	Communication Standard	UVC 1.0.0
	Supported format	YUY2, MJPEG
	Supported MAX Frame rate	60 fps

Software	Supported Resolution	MJPES Output resolutions: 3840*2160 @ 30fps 2688*1944 @ 30fps 2592*1944 @ 30fps 2560*1440 @ 30fps 1920*1080 @ 60fps 1280x720 @ 60fps 960*540 @ 60fps 800x600 @ 60fps 640x480 @ 60fps 640x360 @ 60fps YUY2 Output resolutions: 3840*2160 @ 15fps 2688*1944 @ 20fps 2592*1944 @ 20fps 2560*1440 @ 30fps 1920*1080 @ 30fps 1280x720 @ 60fps 960*540 @ 60fps 800x600 @ 60fps 640x480 @ 60fps 640x360 @ 60fps
	Auto Parameters	White Balance (Manual Option) Exposure (Manual Option)
	Controllable Parameters	Brightness, Contrast, Hue, Saturation, Sharpness, Gamma, White Balance, Backlight Comp, Gain, Exposure, PowerLine Frequency, Low Light Compensation
	Support System	Windows , Linux , Mac Os, Android

4. User Manual Version Descriptions

Version	Description	Date	E-mail
V1.0.0.0		2025.10.30	support@inno-maker.com sales@inno-maker.com

If you have any suggestions, ideas, codes and tools please feel free to email to me. I will update the user manual and record your name and E-mail in list. Look forward to your letter and kindly share.